

Report R6 — Code guide and reference

Tier-1 Krylov sector analysis of quantum cellular automata: data structures, algorithms, and file map

QCA Sector Analysis project (Tier 1)

engine_version 1a.1

Abstract

This report is documentation, not results. It describes the Tier-1 codebase precisely enough to verify every number in R1, R2, and R5 against the source: the exact-arithmetic representation of amplitudes, the construction of the transition graph, the two graph algorithms (union-find for unitary rules, iterative Tarjan plus condensation for dissipative rules), the Tier-1b superoperator and its spectral classification, the fit models and the null-calibrated growth-law extraction, and which Python file is responsible for each of these. Section 2 is the file map; the later sections walk the pipeline stage by stage. Paths are relative to `code/qca_fragmentation/`.

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1 Pipeline at a glance

One rule is a Wolfram number $R \in \{0, \dots, 255\}$; the analysis is run per (rule, N , bc) *unit*, where N is the chain length and $bc \in \{\text{pbc}, \text{obc0}\}$ the boundary convention. The stages are:

Tier 1a — graph. Build the directed one-cycle transition graph on the 2^N computational-basis states and decompose it into strongly connected components (SCCs). For unitary rules the SCCs are Krylov sectors; for dissipative rules the terminal SCCs are recurrent classes (attractors) and the rest are transients. Output: one JSON line per unit in `results/`.

Tier 1b — per-class quantum structure. For each recurrent class build the exact restricted superoperator and read its peripheral spectrum to label the attractor *mixing* / *limit cycle* / *coherence-carrying*. Also the exact channel-level diagnostics (graph faithfulness, Cesàro rank).

Tier 1c — scaling. Aggregate the per- N series, fit growth laws with BIC model selection and exact integer recurrences, and emit CSVs, LaTeX tables, and figures.

The graph stage is *exact* — integers only, no floating point and no tolerances (§4). Floating point enters only in Tier 1b, where a dense eigensolve is unavoidable, and in the Tier-1c regressions.

2 Repository / file map

All code lives under `code/qca_fragmentation/`; data and outputs are siblings of `code/` in the repository root.

path	responsibility
<code>__init__.py</code>	defines <code>ENGINE_VERSION = "1a.1"</code> (stamped into every record; a unit is recomputed only if its stored version differs).
core/ — rule algebra and one-cycle dynamics	
<code>core/rules.py</code>	Wolfram $\leftrightarrow$ tuple $(r_{00}, r_{01}, r_{10}, r_{11})$ encoding; <code>is_unitary</code> , <code>channel_kraus_symbols</code> , reflection / spin-flip maps; HSF oracle numbering.
<code>core/ring.py</code>	exact arithmetic in $\mathbb{Z}[1/\sqrt{2}]$: amplitude $(a + b\sqrt{2})/2^m$, addition, $\times \frac{1}{\sqrt{2}}$, exact-zero, norm.
<code>core/cycle.py</code>	one brickwork cycle; branch propagation; <code>succ(x)</code> (graph edges); <code>one_cycle_branches_labeled</code> (Kraus-labelled, for Tier 1b).
<code>core/ring.py</code> , <code>core/cycle.py</code>	together define the amplitude data structure <code>Branch</code> (§5.1).
graph/ — Tier 1a	
<code>graph/scc.py</code>	<code>analyze</code> (dispatcher), <code>sectors_union_find</code> (unitary), <code>tarjan</code> (iterative, dissipative), <code>_condensation</code> , <code>_analyze_condensation</code> , <code>recurrent_classes</code> ; the <code>GraphResult</code> dataclass.

path	responsibility
quantum/ — Tier 1b	
quantum/peripheral.py	restricted_superoperator, classify_attractor (mixing / limit-cycle / coherent), geometric_multiplicity, graph_faithfulness, full_channel_superoperator, cesaro_rank.
quantum/attractor_structure.py	the structural coherence test: $\dim_{\text{att}}$ vs. basis content, with a spectral-gap guarantee; census.
quantum/dissipative_report.py	assembles the R5 census, Tier-1b table, and Cesàro-gap table.
scaling/ — Tier 1c	
scaling/fits.py	fit_series (M0/M1/M2 BIC), fit_pure_exponential, find_integer_recurrence, find_recurrence_by_period, name_base.
scaling/summary.py	unitary series extraction + per-rule summary + the tab_unitary_summary table.
scaling/dissipative.py	dissipative series; fit_with_period (period selection); exact bases; summarize_rule; CSV + tab_diss_scaling.
scaling/figure.py	unitary figures incl. the growth-base scatter.
scaling/diss_figure.py	dissipative growth-rate scatter maps.
scaling/growth_map.py	the unified (base, α) plane and base-vs-base map over all 256 rules.
scaling/audit.py	coverage gate: missing / interior-gap / too-short units.
scaling/overfit_audit.py	null-model + hold-out calibration of the recurrence and period searches.
drivers and I/O	
results_io.py	JSONL record schema, load_results, has_unit, record_from_graph_result, the checkpoint manifest.
run_rule.py	analyse one unit (CLI); --no-ergodic-abort.
sweep.py	idempotent in-process sweep.
deep_sweep.py	memory-guarded subprocess sweep for large N .
tests/	pytest suite (§12).
outputs (repo root)	
results/{rule}_{bc}.jsonl	one Tier-1a record per N (append-only).
checkpoints/manifest.jsonl	timestamped audit log of completed units.
analytics/	census / audit JSON checkpoints.
figures/, reports/	CSVs, PDFs, and the LaTeX reports.

3 Rule encoding (core/rules.py)

An elementary rule $R = \sum_k c_k 2^k$ is decoded per neighbour pair $(m, n) = (x_{i-1}, x_{i+1})$ into a local channel symbol from the output bits $c_p = c_{4m+n}$ (centre 0) and $c_q = c_{4m+n+2}$ (centre 1):

$$(c_p, c_q) = \begin{cases} (0, 1) \rightarrow \text{I} & \text{identity} \\ (1, 0) \rightarrow \text{V} & \text{Hadamard on the centre} \\ (0, 0) \rightarrow \text{D} & \text{reset to } |0\rangle \\ (1, 1) \rightarrow \text{E} & \text{reset to } |1\rangle \end{cases}$$

So each rule is a 4-tuple $(r_{00}, r_{01}, r_{10}, r_{11}) \in \{\text{I}, \text{V}, \text{D}, \text{E}\}^4$ (wolfram_to_tuple / tuple_to_wolfram, a bijection $4^4 = 256$). Key predicates: `is_unitary(t)` = every symbol in $\{\text{I}, \text{V}\}$ (16 rules); `channel_kraus_symbols(t)` = “has any D/E”, i.e. needs a second Kraus branch. The 16 unitary rules carry an HSF oracle number (wolfram_to_hsf, obc0, V = Hadamard) used only to cross-check against the independent Julia sector counts.

Symmetry maps. `reflect_tuple` swaps $r_{01} \leftrightarrow r_{10}$ (left-right reflection; a valid sector-size reduction for *unitary* rules only, since the even/odd brickwork order matters for dissipative graphs). `spinflip_wolfram` implements $0 \leftrightarrow 1$ conjugation, valid only for V-free rules (Hadamard breaks it: $XHX \neq \pm H$).

4 Exact amplitudes: the ring $\mathbb{Z}[1/\sqrt{2}]$ (`core/ring.py`)

Every amplitude produced by an I/V/D/E circuit lies in $\mathbb{Z}[1/\sqrt{2}]$. It is stored as

$$\text{value} = \frac{a + b\sqrt{2}}{2^m}, \quad a, b \in \mathbb{Z} \text{ (arbitrary precision),}$$

with a *single* exponent m shared by a whole branch dictionary. The one non-trivial gate, a conditional Hadamard, multiplies fired components by $1/\sqrt{2}$; with $(a+b\sqrt{2})/\sqrt{2} = (2b+a\sqrt{2})/2$ this stays integer-closed:

operation	effect on numerator (a, b) , exponent m
identity / value unchanged, $m \rightarrow m+1$	$(a, b) \rightarrow (2a, 2b)$ (<code>double</code>)
$\times 1/\sqrt{2}$, $m \rightarrow m+1$	$(a, b) \rightarrow (2b, a)$ (<code>div_sqrt2</code>)
exact zero test	$a = 0 \wedge b = 0$ (<code>is_zero</code> ; no tolerance)
squared norm over 4^m	$ a + b\sqrt{2} ^2 = (a^2 + 2b^2) + (2ab)\sqrt{2}$ (<code>prob_int</code>)

Because $\sqrt{2}$ is irrational, $a + b\sqrt{2} = 0$ with integer a, b forces $a = b = 0$: the zero test is exact and the graph never needs a numerical threshold. `renorm` halves all numerators while both parities are even (keeping the integers bounded); `branch_norm` returns the squared norm as two rationals (rational part, $\sqrt{2}$ part) — the $\sqrt{2}$ part must vanish for any physical total, and equals 1 exactly for a normalised unitary branch (a runtime check).

5 One cycle and the transition graph (`core/cycle.py`)

5.1 The Branch data structure

$$\text{Branch} = (\{ \text{bitmask} \rightarrow (a, b) \}, m)$$

a dict from computational-basis state (an `int` bitmask, site $i = \text{bit } i$) to its numerator (a, b) , plus the shared exponent m . One cycle applies the local update at all *even* sites $0, 2, 4, \dots$ first, then all *odd* sites (brickwork); odd-layer controls read the post-even-layer state. A rule is *compiled* once per (N, bc) (`_compile`, `lru_cached`) into an ordered list of site steps with precomputed neighbour bit-positions (-1 marks a frozen-0 obc0 edge).

5.2 Kraus branching

A reset (D/E) splits a branch into two: the *no-jump* A_0 sub-branch and the *jump* A_1 sub-branch (`_split_site` returns `(a0, a1)`). Branches are **never merged** — merging would fabricate interference between incoherent Kraus alternatives. Two propagators:

- `one_cycle_branches(x, N, t, bc)` — surviving branches after one cycle. Unitary rules take a single-branch fast path (`_apply_site_unitary`).
- `one_cycle_branches_labeled` — additionally tags each branch with the `frozenset` of sites where A_1 fired. That set uniquely names the global Kraus operator $K_b = \prod_i A_{b_i}(i)$ in the fixed processing order, so branches of different inputs can be paired by label to assemble the exact superoperator in Tier 1b.

5.3 The edge relation

$$\text{succ}(x) = \{y : \langle y | \Phi_C(|x\rangle\langle x|) | y \rangle \neq 0\} = \bigcup_{\text{branches}} \text{supp}(\text{branch}), \quad (1)$$

the union of branch supports after one cycle, with exact-zero pruning. This is the directed edge set of the transition graph; it is streamed, never materialised. Equivalently `succ` is the support of the dephased kernel $M_{yx} = \sum_{\mu} |\langle y | K_{\mu} | x \rangle|^2$ — exactly the QCA *monitored* in the computational basis after every cycle (see R5 for why this differs from the unmonitored channel for V-carrying rules).

6 Tier 1a: graph decomposition (graph/scc.py)

`analyze(rule, N, bc, t)` returns a `GraphResult` and dispatches on unitarity.

6.1 Unitary rules: union-find

A unitary (doubly stochastic) channel makes every state recurrent, so each weakly connected component is a single SCC and the sector partition equals the undirected connected components. `sectors_union_find` streams the edges once with union by size and path compression — $O(2^N)$ integers, no stored adjacency, no DFS stack. This is the memory-light path that reaches $N = 22$ (2 GB $\rightarrow$ 45 MB at $N = 17$ versus stack-Tarjan). It early-exits to *ergodic* when a component first exceeds $f_{\text{erg}} \cdot 2^N$.

6.2 Dissipative rules: iterative Tarjan + condensation

The directed graph is decomposed by an **iterative** (explicit-stack) Tarjan (`tarjan`) — recursion would overflow at $N \gtrsim 20$. It returns `comp_id` (per-node SCC index, numbered in reverse-topological order) and the component sizes, and early-exits to *ergodic* if any SCC exceeds $f_{\text{erg}} \cdot 2^N$ or the node count exceeds `node_budget`. Two further streaming passes:

- `_condensation` — builds the successor sets of the condensation DAG (one entry per SCC).
- `_analyze_condensation` — *terminal* SCCs (no outgoing edge) are the recurrent classes; a Kahn topological sort gives the longest path (= `transient_depth`); a reverse-topological DP assigns each transient a single reachable terminal or a “multiple” flag, from which the exclusive `basin` sizes and the `shared_basin_size` follow.

`recurrent_classes` returns the member-state lists of the terminal SCCs and is the entry point Tier 1b consumes.

6.3 The GraphResult / record fields

`record_from_graph_result` (`results_io.py`) serialises to one JSON line. The schema (field order in `results_io.FIELDS`):

field	meaning
<code>rule, bc, N</code>	the unit key
<code>n_scc</code>	number of SCCs
<code>n_recurrent</code>	number of terminal SCCs (sectors / attractors)
<code>sizes_recurrent</code>	terminal-SCC sizes, descending (capped at 2048)
<code>sizes_scc</code>	all SCC sizes, descending (capped)
<code>size_hist</code>	{size:count} histogram (exact reconstruction if capped)
<code>sizes_truncated</code>	true if >2048 explicit sizes were dropped
<code>sizes_basins</code>	exclusive basin sizes, descending
<code>shared_basin_size</code>	states whose forward orbit reaches > 1 terminal
<code>transient_depth</code>	longest condensation-DAG path (relaxation proxy)
<code>n_transient_scc</code>	number of non-terminal SCCs
<code>ergodic_flag, ergodic_bound</code>	set when early-exit fired; the SCC size
<code>attractor_types, d_max_quantum</code>	filled by Tier 1b (else null)
<code>runtime, engine_version</code>	wall seconds; "1a.1"

`sizes_from_record` reconstructs the full multiset from `size_hist` when the explicit list was capped, so the sorted sizes ($D_{\max}$ etc.) are never corrupted by the cap. `load_results` returns $\{N : \text{record}\}$; `has_unit` gates idempotency on `engine_version`.

7 Tier 1b: per-class quantum structure (quantum/peripheral.py)

For a recurrent class R (a terminal SCC, closed under the channel) the exact restricted superoperator on $\text{span}(R) \otimes \text{span}(R)^*$ is assembled from the labelled Kraus branches (§5):

$$\Phi_R[(y, y'), (x, x')] = \sum_b \langle y | K_b | x \rangle \overline{\langle y' | K_b | x' \rangle}, \quad (2)$$

a dense $|R|^2 \times |R|^2$ real matrix (`restricted_superoperator`); matrix elements are exact $\mathbb{Z}[1/\sqrt{2}]$ values cast to float only here. Classification (`classify_attractor`) reads the *peripheral* spectrum ($|\lambda| > 1 - 10^{-8}$):

condition	label
unique $\lambda = 1$, no other peripheral eigenvalue	mixing (pointer / dark state)
unique $\lambda = 1$, peripheral roots of unity, eigenoperators diagonal	limit cycle (period q)
geometric mult. of $\lambda = 1$ exceeds # diagonal fixed operators	coherence-carrying ($d_k = \sqrt{\text{mult}}$)

The crucial subtlety is `geometric_multiplicity` (via SVD of $\Phi_R - \lambda I$): the *algebraic* multiplicity is inflated by the transient $\rightarrow$ recurrent Jordan structure and must not be used. Classes larger than `max_size` = 32 are reported *unresolved* (the dense eigensolve is $O(|R|^4)$).

Channel-level diagnostics (validation, $N \lesssim 6$). `full_channel_superoperator` builds the dense $4^N \times 4^N$ channel from per-site Kraus $\{A_0, A_1\}$ as $\prod_{\text{sites}} (A_0 \otimes \bar{A}_0 + A_1 \otimes \bar{A}_1)$. `cesaro_rank` is the geometric multiplicity of its eigenvalue 1 (the protected-coherence dimension). `graph_faithfulness` returns `leak_fix` ($\text{Fix}(\Phi)$ weight outside the graph's recurrent block) and `leak_flow` (population off the recurrent set after many cycles); a rule is *faithful* iff both vanish.

Structural coherence test (quantum/attractor_structure.py). `graph_faithfulness` catches only the weaker failure. The structural test computes $\dim_{\text{att}} = \dim \text{supp}$ of the Cesàro limit of $\Phi^n(\mathcal{K}/d)$ and the number of basis states inside it; the excess `coh_dirs` = $\dim_{\text{att}} - \text{basis_in_att} > 0$ is exactly the structure no basis-state graph can represent. It is iterated to a genuine *spectral gap* (smallest kept eigenvalue over largest dropped $> \text{GAP_MIN} = 10^4$), not a fixed burn-in — burn-in 600 gives rule 19 a bogus $\dim_{\text{att}} = 35$ at $N = 6$; converged it is 5. `census` runs this over all dissipative rules and checkpoints to `analytics/`.

8 Tier 1c: fit models (scaling/fits.py)

8.1 BIC model selection: fit_series

On the exact $\ln y$ values, three nested models are fitted by least squares and selected by BIC:

$$\text{M0: } \ln y = c; \quad \text{M1: } \ln y = c + \alpha \ln N; \quad \text{M2: } \ln y = c + \alpha \ln N + \kappa N.$$

$\text{BIC} = n \ln(\text{RSS}/n) + k \ln n$, with a relative RSS floor so that a near-perfect fit does not let floating-point noise dominate $n \ln \text{RSS}$ (then the fewest-parameter model wins on the $k \ln n$ penalty). The class label: M0 $\rightarrow$ *constant*, M1 $\rightarrow$ *polynomial*, M2 $\rightarrow$ *exponential* iff $|\kappa| > \text{_KAPPA_EPS} = 0.02$ (a linear sector count yields a spurious $\kappa \sim 0.008$ that must not read as exponential). The $\alpha \ln N$ term is *always* carried when reporting κ , because binomial / charge sectors carry $\sqrt{N}$ prefactors ($\alpha \approx -\frac{1}{2}$) that otherwise bias κ ; κ is returned with a leave-one-out spread.

8.2 Pure-exponential rate: fit_pure_exponential

The M2 $\alpha \ln N$ term is unstable on the ragged *dissipative* series (rule 90's $D_{\max} = 1, 15, 1, 7, 3, 63, 2, 127$ gives $\alpha = -7.5$ and an impossible base 11). For rate extraction on those, the two-parameter $\ln y = c + \kappa N$ is used instead; it carries a **bounded** flag ($\kappa \leq \ln 2$, since $D_{\max} \leq 2^N$). The report's rule of thumb (R2): read the base of a *room-packing* (geometric b^N) series from this two-parameter fit, and the base of a *binomial* series from M2 — the mechanism, not the residual, selects the estimator.

8.3 Exact integer recurrences: find_integer_recurrence

Searches for the smallest-order linear recurrence $a(n) = \sum_i c_i a(n-i)$, $c_i \in \mathbb{Z}$: the first k equations are solved over $\mathbb{Q}$ (**Fraction**), integrality is required, and the candidate is then *verified on every remaining term*. The growth base is $\max |\text{root}|^{1/\text{step}}$ of the characteristic polynomial — an algebraic number, not a regression. Guards: a repeated unit root (a polynomial series) is snapped to base 1 (**np.roots** splits $(x-1)^3$ into a cluster at 1.000007); **step** is the N -spacing (2 for a parity subsequence, so the root is taken to the $1/\text{step}$ power); **max_skip** may step over a finite-size transient but is *off by default* (it costs false positives — §11). Matched constants are named by **name_base** against **_NAMED_BASES** (φ , $\sqrt{3}$, plastic ρ , supergolden, tribonacci, $4^{1/5}$, the parity-doubled surds, ...).

8.4 Period splitting: find_recurrence_by_period

Many dissipative series oscillate with N because the attractor is a spatial pattern that only fits a commensurate ring (parity is the common case: an odd **pb**c ring cannot host the two Néel states). **find_recurrence_by_period** runs the recurrence search per residue class $N \bmod p$ and accepts only when *every* class has one and they agree on the base.

9 Tier 1c: scaling drivers

9.1 Unitary — scaling/summary.py

load_series reads **results/** and builds the per- N series **n_recurrent** (sectors), **d_max** (largest sector), **max_basin**; ergodic-flagged units are excluded and the first ergodic N recorded. **summarize_rule** runs **fit_series** on each and **latex_summary_table** writes **tab_unitary_summary.tex**.

9.2 Dissipative — `scaling/dissipative.py`

The dissipative series are the same four keys but mean attractor count, largest attractor, largest basin, and transient depth. The extra machinery:

- `fit_with_period` chooses a commensurability period by **BIC over the pooled per-class log-residuals**, requiring a decisive margin (`_BIC_MARGIN= 10`, Kass–Raftery “very strong”), at least `_MIN_PER_CLASS= 4` points per class, and the precondition that the joint fit is actually failing (`_RESID_BAD = 0.15`). An exact per-class recurrence is accepted as an independent second route. This replaced an earlier ratio-of-residuals rule that over-split (§11).
- `summarize_rule` emits, per series, the growth class, base (exact where a recurrence exists), period, and — for *irregular* series — the `irregular_kind`: *arithmetic* (a **V**-free rule is deterministic on basis states, so $D_{\max}$ is a cycle length / multiplicative order, not a growth law — e.g. rule 90 over GF(2)) vs. *undetermined* (a **V**-carrying rule whose period the sizes cannot settle).

`attractor_class` gives the coarse graph structure (unique-point / multistable / extended / multi-extended).

9.3 Figures

`scaling/figure.py` plots $\ln y$ vs N and the unitary growth-*base* scatter; `scaling/diss_figure.py` the dissipative growth-rate maps; `scaling/growth_map.py` the unified maps over all 256 rules — the (base, α) plane (leading rate vs sub-leading power, non-degenerate across exponential / polynomial / ergodic) and the base-vs-base view. A filled marker is an analytically exact base; an open marker is a fit.

10 Sweep drivers and idempotency

`run_rule.run_unit` computes one unit unless it is already cached for the current `engine_version` (`has_unit`); results are append-only, so a sweep is safe to kill and restart. `--no-ergodic-abort` performs the full decomposition even past the ergodic threshold, keeping the flag (used to prove the “ergodic” unitary rules are N -independent; see R2).

`sweep.py` walks (rule, N , bc) in-process and stops enlarging N once a rule is ergodic or budget-capped (fragmentation cannot reappear at larger N). `deep_sweep.py` runs each unit in a *subprocess* under an address-space `RLIMIT_AS` cap and a wall timeout, in three cost tiers (small / large / huge), so one runaway unit dies alone instead of OOM-ing the night; N is walked upward so an interrupted run still leaves a hole-free dataset at every completed size. The genuine ceiling on 32 GB is $N = 17$ for the dissipative rules ($N = 16$ for the three-Hadamard rules 19/55, whose directed Tarjan does not fit 2^{17} states).

11 Audits and overfitting calibration (`scaling/`)

`audit.py` is the coverage gate: it reports MISSING (no data; expected only for ergodic units), interior GAPS (an absent N inside the range — fatal to a fit, non-zero exit), SHORT (fewer points than a recurrence of a given order needs, $2k+1$), and the parity-effective length. `overfit.audit.py` calibrates the two searches that could fake structure:

- **null recurrence rate** — on structureless surrogates the exact recurrence search fires 0/4000; this is what licenses it to certify periods BIC cannot afford.

- **null period rate** — on smooth exponentials +25% noise the calibrated period split fires $\sim 3\%$, versus 21% for the old ratio-of-residuals rule and 33% for plain argmin-BIC.
- **hold-out extrapolation** — refit on all but the largest three sizes and predict them: 854/854 testable series extrapolate exactly, which is far stronger evidence for the algebraic bases than any in-sample residual.

12 Tests (tests/)

file	covers
test_rules.py	encoding bijection, unitarity, symmetry maps, HSF table
test_ring.py	$\mathbb{Z}[1/\sqrt{2}]$ closure, exact zero, norms
test_cycle.py	branch propagation, succ , unitary single-branch norm
test_scc.py	Tarjan / union-find agreement, condensation, ergodic exit
test_peripheral.py	superoperator vs dense channel, attractor labels
test_attractor_structure.py	rule-27 coherence, gap-driven convergence
test_fits.py	recurrences, φ , parity bases, $4^{1/5}$ near-degeneracy
test_diss_scaling.py	period selection, null false-split, rule-90 GF(2)
test_growth_map.py	anchor-rule (class, base, α)
test_deep_sweep.py	ergodic-skip, cost ordering, tier table

13 Reproducing the pipeline from scratch

All commands use `~/pyenvs/main/bin/python` from `code/`.

1. **One unit:**

```
python -m qca_fragmentation.run_rule --rule 156 --N 6-18 --bc obc0
```
2. **Sweep** (idempotent; resume by re-running):

```
python -m qca_fragmentation.sweep --which unitary --n-min 6 --n-max 18
```

```
python -m qca_fragmentation.deep_sweep --n-min 14 --n-max 17 --which dissipative
```
3. **Coverage gate** (non-zero exit on gaps):

```
python -m qca_fragmentation.scaling.audit
```
4. **Scaling tables / CSV:**

```
python -m qca_fragmentation.scaling.summary (unitary)
```

```
python -m qca_fragmentation.scaling.dissipative (dissipative)
```
5. **Figures:**

```
python -m qca_fragmentation.scaling.figure
```

```
python -m qca_fragmentation.scaling.diss_figure
```

```
python -m qca_fragmentation.scaling.growth_map
```
6. **Overfitting calibration:**

```
python -m qca_fragmentation.scaling.overfit_audit
```
7. **Tier 1d — pair graph, certificates, weak charges** (separate `results_tier1d/` store, `tier1d_version 1d.1`; does *not* recompute Tier 1a):

```
python -m qca_fragmentation.run_rule --rule 22 --N 4-12 --bc pbc --tier 1d
```

```
python -m qca_fragmentation.pair_sweep --which coherent --n-min 4 --n-max 12
```

```
python -m qca_fragmentation.quantum.weak_charges --core
```

```
python -m qca_fragmentation.scaling.pair_scaling --bc pbc (sandwich table)
```

```
python -m qca_fragmentation.scaling.pair_figure --bc pbc (coverage / sandwich / parity)
```

8. **Tests:** `python -m pytest qca_fragmentation/tests -q`

Every unit is cached in `results/` and logged in `checkpoints/manifest.jsonl`; nothing recomputes unless `engine_version` changes or `--force` is given.